

A deep learning framework for portfolio performance in
commodity market

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ABSTRACT

This paper presents a novel approach to portfolio optimization explicitly tailored for the dynamic and volatile environment of commodity markets. Leveraging historical data from 2014 to 2023 across five distinct commodity categories—Metals, Softs, Energy, Grains, and Meats—the study aims to devise an optimized portfolio strategy from January to February 2024 for all the five commodities. The objective is to maximize returns while minimizing volatility, thus enhancing investor profitability and risk management.

Our methodology begins with a unique forecasting framework, integrating CNN and LSTM layers to predict average returns for the upcoming two-month period. This predictive model serves as the foundation for subsequent optimization strategies. If the forecasted returns of any category is positive, we individually employ Hierarchical Risk Parity (HRP) and Genetic Algorithms (GA) to determine optimal weight allocations. In instances of predicted negative returns, short-selling strategy is implemented. Equal weighting across commodities is employed to mitigate bias, and an optimization function is used that minimizes the product of weights and volatility for each commodity. Weight adjustments are constrained within a predefined range ensuring portfolio diversification while maintaining balance.

The core innovation lies in developing an optimization function that strategically balances the trade-off between return potential and volatility. This function minimizes the weighted volatility of individual commodities while ensuring that the sum of all weights equals unity. Additionally, each weight can vary within a specified neighborhood, providing flexibility and adaptability to changing market conditions. Comparative analysis against established portfolio optimization methodologies such as Modern Portfolio Theory (MPT), HRP, and HERC demonstrates our proposed approach's superior performance and robustness. In summary, this study contributes to the ongoing discussion on portfolio optimisation by offering a comprehensive strategy created especially for commodities markets.

LIST OF SYMBOLS OR ABBREVIATIONS

LSTM	Long Short Term Memory
HERC	Hierarchical Equal Risk Contribution
HRP	Hierarchical Risk Parity
CNN	Convolutional Neural Network
LSTM	Long Short Term Memory
MPT	Modern Portfolio Theory
GA	Genetic Algorithm
RNN	Recurrent Neural Network

LIST OF FIGURES

3.1	metals equity progression	7
3.2	grains equity progression	8
3.3	softs equity progression	8
3.4	meats equity progression	9
3.5	energy equity progression	9
3.6	configuration	18
3.7	model summary	19
3.8	Training	23
3.9	Testing	23
3.10	train loss and val loss	24
3.11	train mse and val mse	24
3.12	train mae and val mae	24
3.13	Average predicted returns for Metals over the next 2 months.	25
3.14	Average predicted returns for Energy over the next 2 months.	26
3.15	Average predicted returns for Softs over the next 2 months.	26
3.16	Average predicted returns for Meats over the next 2 months.	27
3.17	Average predicted returns for Grains over the next 2 months.	27
4.1	Cumulative Return ENERGY	29
4.2	Cumulative Return SOFTS	30
4.3	Cumulative Return MEATS	31

Contents

Certificate	i
Academic Integrity and Copyright Disclaimer	ii
Abstract	iii
List of Symbols or Abbreviations	iv
List of Figures	v
1 Introduction	1
1.1 Introduction	1
2 Literature Review	4
2.1 Related Work	4
3 Experimental Setup	6
3.1 Dataset	6
3.2 Models and Methods	10
3.2.1 CNN	10
3.2.2 LSTM	10
3.2.3 PORTFOLIO MODEL	10
3.2.4 PROPOSED METHOD	13
4 Results and Discussion	28
4.1 Results	28

4.1.1	<u>COMMODITIES GIVING POSITIVE AVERAGE RETURNS</u>	28
4.1.2	<u>COMMODITIES GIVING NEGATIVE AVERAGE RETURNS</u>	32
5	Conclusion	33
6	Limitations and Future Work	34
7	Acknowledgment	35
	Bibliography	35

Chapter 1

Introduction

1.1 Introduction

Portfolio optimization in commodity markets presents unique challenges due to the dynamic nature of these markets and the inherent volatility of commodity prices. This thesis focuses on developing a comprehensive portfolio optimization framework for commodity investments. Our study encompasses historical data from 2014 to 2023 across five distinct commodity categories: metals, softs, energy, grains, and meats.

The main goal of this research is to identify optimal portfolio weights that maximize returns and minimize volatility from January to February 2024 for all five commodities. This timeframe serves as a crucial test period, allowing for the evaluation of the effectiveness of our proposed methodology in a real-world scenario.

We propose a novel approach that integrates advanced forecasting techniques with cutting-edge optimization algorithms to achieve this objective. The first step of our methodology involves utilizing a hybrid model comprising CNN and LSTM layers to predict the average returns for the upcoming two-month period.[1],[2]

Subsequently, we employ a multi-step optimization process. When the predicted returns for the target period are positive, we apply the hierarchical risk parity (HRP)[3] and genetic algorithm (GA)[4] individually to determine the optimal

weight allocations for each commodity. The optimization function is then used to minimize the product of weights and volatility for each commodity while ensuring that the sum of all weights equals unity. Additionally, we introduce a parameter that allows for weight variations within a specified neighborhood, enhancing the adaptability of the portfolio to changing market conditions.[5]

In cases where negative returns are predicted for any segment, a short-selling strategy is implemented, ensuring flexibility and risk mitigation in adverse market scenarios. This strategy allows us to capitalize on downward price movements in specific commodity segments, potentially offsetting losses and enhancing portfolio performance. We assign equal weight to all to maintain fairness and objectivity in our analysis, thereby mitigating bias in the portfolio construction process. Each commodity is given an equal share of the portfolio, regardless of its individual characteristics or historical performance, ensuring that our methodology remains impartial and systematic.

We employ an optimization function to balance the trade-off between return potential and volatility in optimizing the portfolio weights. This function minimizes the product of weights and volatility for each commodity while ensuring that the sum of all weights equals unity. Additionally, we introduce a parameter that allows for weight variations within a specified neighborhood, providing flexibility and adaptability to changing market conditions. By incorporating this optimization function, we wish to build a portfolio that maximizes returns while minimizing risk, thereby enhancing investor profitability and risk management in commodity markets.

Throughout this study, we conduct a comparative analysis against established portfolio optimization methodologies, including Modern Portfolio Theory (MPT)[6], HRP, and HERC[7]. By evaluating the performance and robustness of our proposed approach against these benchmarks, we aim to demonstrate its superiority in enhancing risk-adjusted returns.

In conclusion, this research adds to the current conversation on portfolio optimization by introducing a comprehensive methodology tailored specifically for commodity investments. By leveraging advanced forecasting techniques and optimization

algorithms, our approach seeks to provide investors with a powerful tool for navigating the complexities of commodity markets and achieving their investment objectives.

Chapter 2

Literature Review

2.1 Related Work

The traditional mean-variance portfolio theory contributed to the financial perspective's appreciation of diversity. Because the expected return on an investment is a crucial component of portfolio optimization, choosing assets in advance is essential for portfolio management. With high-quality asset input, applying complex portfolio optimization methods is beneficial. The application of AI and machine learning to predict stock market returns and volatility has been the subject of extensive research. This is because a wealth of data is available, and computers have superior processing capabilities. For learning and retaining financial time-series data over an extended period of time, early statistical and machine learning techniques are not the best. [8], [9].

The ability of deep learning to develop complex self-adaptation and nonlinear mapping mechanisms gave rise to the field. LSTM neural networks were used by Fischer and Krauss [10] to predict the direction of the component stocks of the SP 500 between 1992 and 2015. It was demonstrated that LSTM neural network-based portfolio models outperformed previous models.

A CNN-based deep learning method for forecasting the movement of Chinese stock market values was proposed by Chen and He [11]. The internet-sourced open price,

high price, low price, close price, and volume of stock were the inputs used by the structure. The results demonstrate the ability of a CNN-based deep learning system to forecast changes in Chinese stock values with high accuracy.

A CNN was trained using images of 2D technical candlestick charts by Komori [12] to develop a trading strategy. Ta et al. [13] used the MV model, equal-weighted approach, Monte Carlo simulation, and LSTM neural networks to construct portfolios. The research' findings showed that LSTM outperformed linear regression and support vector machines in terms of prediction accuracy. Using an LSTM architecture, S. Bhandari et al. [14] forecasted the closing price of the SP 500 index the next day. The experiment results show single-layer LSTM provides greater prediction accuracy and a better fit than the multilayer LSTM models.

The SK-GCN model, which divides equities into many categories using a graph convolutional neural network, was created by Xu et al. [15]. This model employs brief text classification and external nodes to classify stocks using two convolutional layers and activation functions. To build the dataset, they crawled every stock listed on the Oriental Fortune website's GEM. Under limited sample training, they achieved an accuracy of 83.04.

Chapter 3

Experimental Setup

3.1 Dataset

The dataset utilized in this thesis encompasses daily price data for five distinct commodity categories, namely **Metals**, **Energy**, **Softs**, **Grains**, and **Meats** , spanning from June 2014 to December 2023 for training and January to February 2024 for testing . The dataset is meticulously collected from **Investing.com**. Data collection involves accessing daily price quotes for each commodity from Investing.com's historical data archives and organizing them into a structured dataset. For using CNN-LSTM model Open, Close, High and Low data for each of the commodity is being used . The dataset is curated to ensure accuracy and consistency, allowing for reliable analysis and modeling . Investing.com's extensive coverage and data quality make it a reliable source for conducting empirical research.

Below is the **Equity Progression** for all commodities. Equity progression refers to the historical evolution of the value of investments in a particular asset or group of assets over a specified period.

Metals Segment: The historical data includes prices for **Aluminium**, **Lead**, **Palladium**, **Tin**, **Zinc**, **Gold**, **Silver**, and **Nickel**. These metals represent critical components of the global metals market and are essential for various industrial and investment purposes.

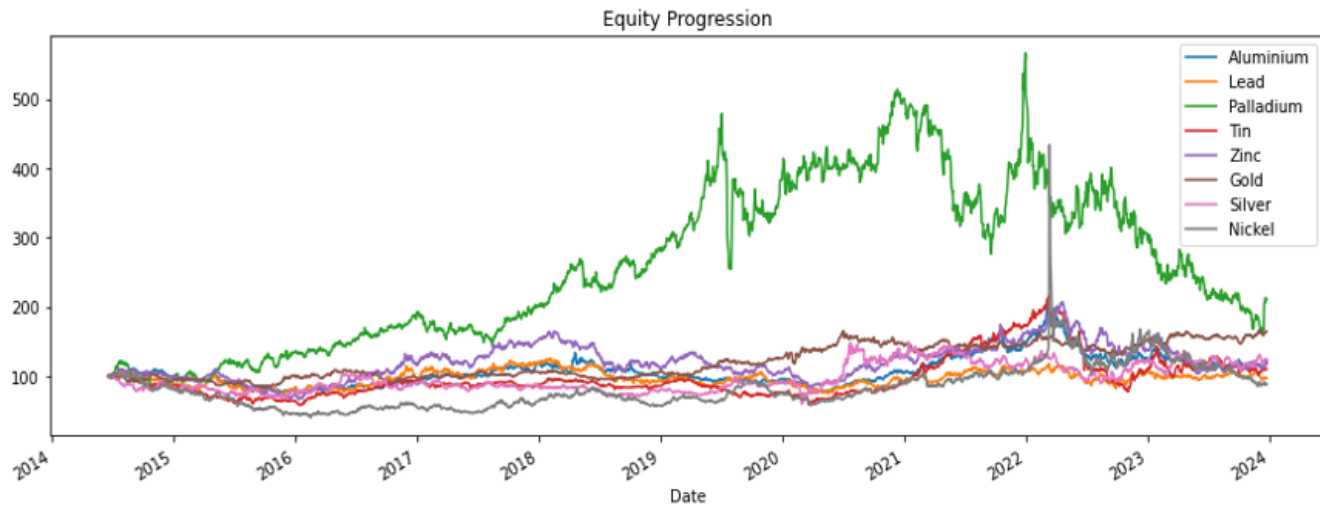


Figure 3.1: metals equity progression

Grains Segment: The Grains segment comprises historical price data for **Oats**, **Rough Rice**, **US Corn**, **US Soybean Meal**, **US Soybean Oil**, **US Soybean**, and **US Wheat**. These commodities are crucial in the agricultural sector and fundamental to global food production.

Softs Segment: The Softs segment encompasses historical price data for **Lumber**, **Orange**, **Cocoa**, **US Coffee C**, **US Cotton**, and **US Sugar**. These soft commodities are derived from agricultural products and find applications in various industries, including food and textile manufacturing.

Meats Segment: The historical data for Meats includes **Feeder Cattle**, **Lean Hogs**, and **Live Cattle** prices. These commodities represent livestock products integral to the global meat industry and are subject to fluctuations influenced by demand, supply, and market sentiment.

Energy Segment: The Energy segment includes historical price data for **Crude Oil**, **Heating Oil**, **Gasoline RBOB**, and **Natural Gas**. These energy commodities are essential for powering various sectors of the economy, including transportation, manufacturing, and heating.

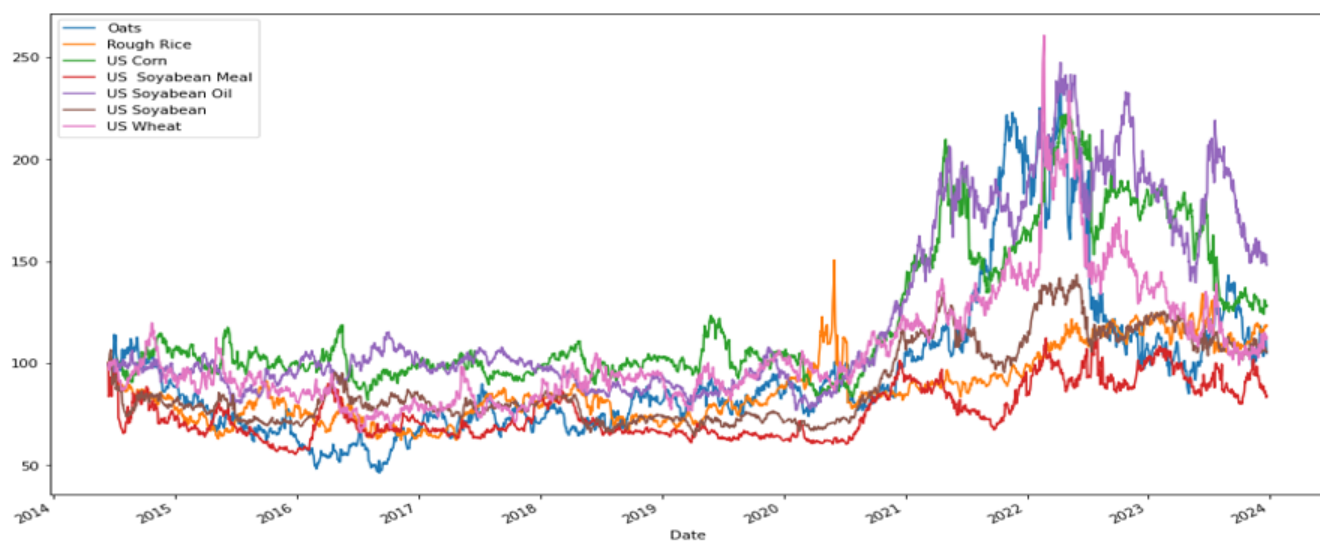


Figure 3.2: grains equity progression

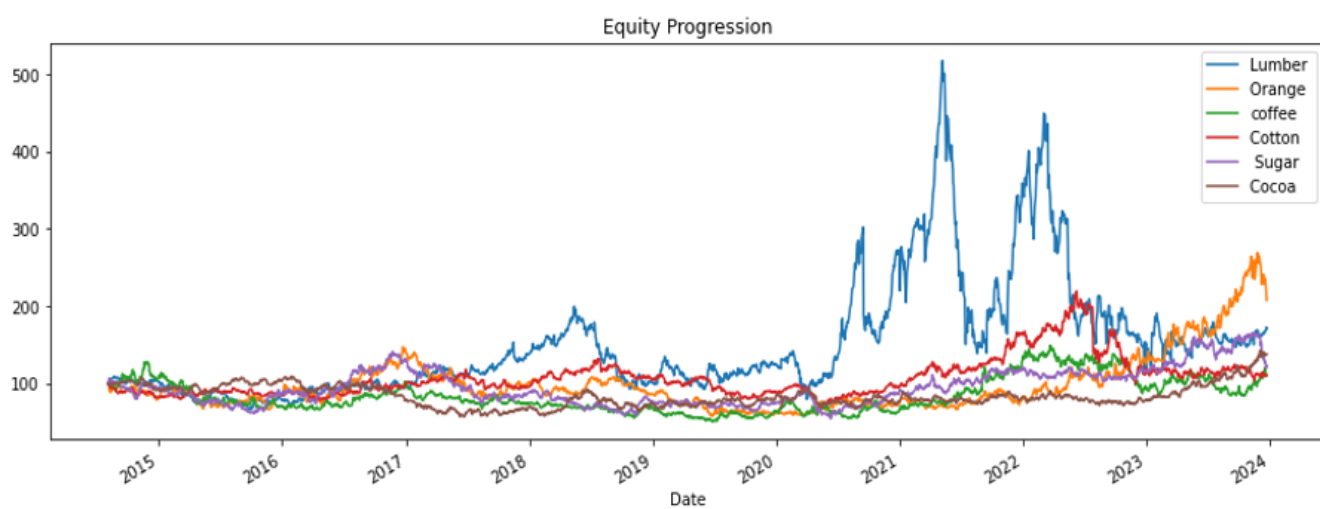


Figure 3.3: softs equity progression

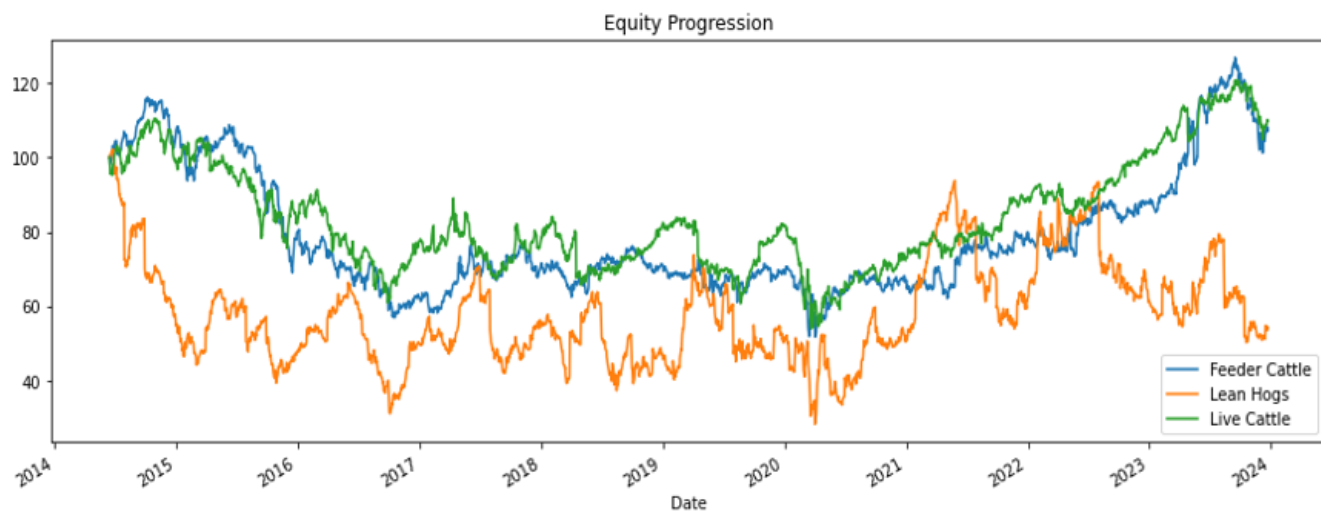


Figure 3.4: meats equity progression

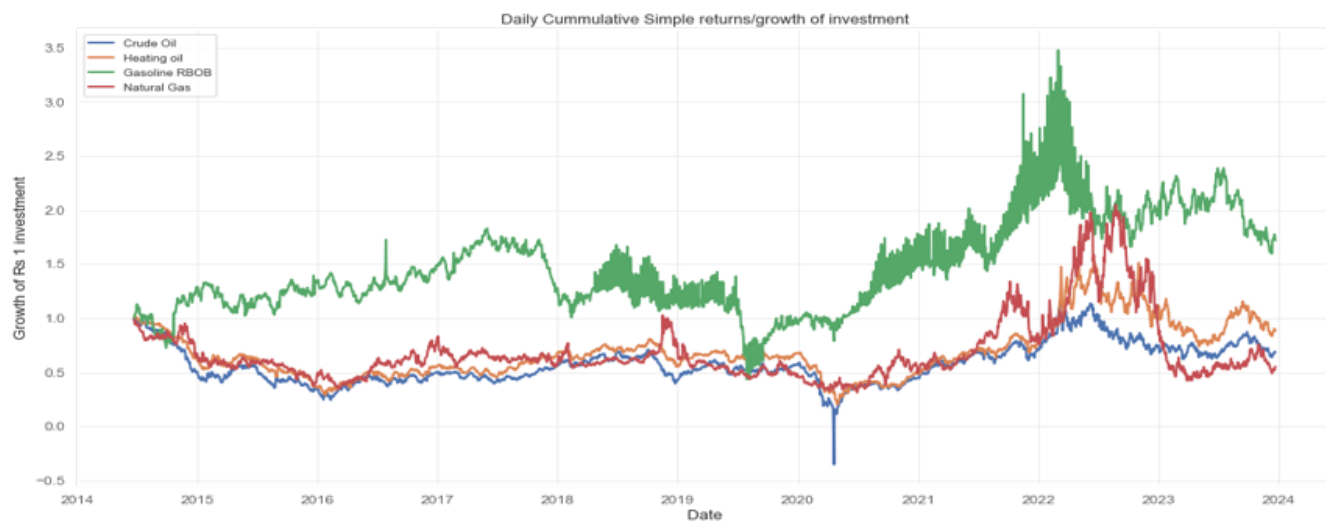


Figure 3.5: energy equity progression

3.2 Models and Methods

3.2.1 CNN

CNN was proposed by Lecun et.al[16]. Convolutional layers apply filters (kernels) to input images, extracting features such as edges, textures, and patterns. These filters slide over the input image, performing convolutions to produce feature maps. CNNs typically conclude with a softmax layer for classification tasks, which assigns probabilities to different classes. CNNs can effectively identify complex and nonlinear patterns.

3.2.2 LSTM

Long Short-Term Memory (LSTM)[17] networks are specialized form of RNN tailored to grasp intricate patterns within sequential data, especially in contexts like financial time series analysis. Unlike traditional RNNs, LSTMs have memory cells that retain information over extended sequences. This retention capability stems from incorporating three fundamental components: input gates, forget gates, and output gates. These gates regulate the flow of information into, out of, and within the memory cells, enabling the network to update and utilize information over time selectively. During training, LSTM networks employ backpropagation through time (BPTT), a variant of the standard backpropagation algorithm that extends across multiple time steps, to adjust their parameters and optimize their performance. Due to their adeptness at capturing long-term dependencies, LSTMs find widespread application across various domains, including natural language processing, speech recognition, and financial forecasting. In finance, LSTMs excel at discerning intricate patterns within market data, enabling more accurate predictions and informed decision-making.

3.2.3 PORTFOLIO MODEL

HRP

[3]

1. **Correlation Matrix:** Compute the $N \times N$ correlation matrix C from the N asset returns. This matrix represents the pairwise correlations between assets.
2. **Hierarchical Clustering:** Use hierarchical clustering to group assets based on their pairwise correlations, resulting in a dendrogram.
3. **Inverse Variance Calculation:** Compute the inverse variance of each cluster. Let Σ_i denote the covariance matrix of cluster i , and v_i be the inverse variance of cluster i , calculated as the reciprocal of the sum of variances of assets within the cluster:

$$v_i = \frac{1}{\text{Trace}(\Sigma_i)}$$

4. **Recursive Bisection:** Recursively bisect the dendrogram to form a hierarchical risk parity tree. At each step, select the cluster with the highest inverse variance and divide it into two sub-clusters until all assets are assigned to individual clusters.
5. **Weights Calculation:** Compute the weights for each asset based on the hierarchical risk parity tree. Let w_{ij} be the weight of asset j in cluster i . The weight is determined by the inverse variance of the cluster and the correlation between the asset and the other assets in the cluster. Let ρ_{kj} be the correlation between asset j and asset k in cluster i . Then, the weight w_{ij} is calculated as:

$$w_{ij} = \frac{v_i}{\sum_{k \in \text{Cluster}(i)} \frac{v_k}{v_i} \rho_{kj}}$$

6. **Portfolio Construction:** Construct the optimal portfolio by allocating capital to each asset based on the computed weights. The resulting portfolio is designed to achieve risk parity, where each asset contributes equally to the portfolio's overall risk.

Genetic Algorithm

[4]

Let P be the population of portfolios, where each portfolio p_i consists of N assets

with corresponding weights w_{ij} , $i = 1, 2, \dots, |P|$ and $j = 1, 2, \dots, N$.

1. **Initialization:** Initialize the population P with random or heuristic portfolio allocations.
2. **Fitness Evaluation:** Evaluate the fitness of each portfolio p_i in the population using a fitness function $f(p_i)$, which measures the performance based on predefined criteria:

$$f(p_i) = \text{Fitness}(p_i)$$

3. **Selection:** Select portfolios from the population to create the mating pool for the next generation. The probability $P(p_i)$ of selecting portfolio p_i is proportional to its fitness:

$$P(p_i) = \frac{f(p_i)}{\sum_{j=1}^{|P|} f(p_j)}$$

4. **Crossover:** Perform crossover on pairs of selected portfolios to create offspring. Let p_{parent1} and p_{parent2} be the selected parent portfolios. The crossover operation generates two offspring portfolios $p_{\text{offspring1}}$ and $p_{\text{offspring2}}$ by exchanging genetic material (asset weights) between the parents.
5. **Mutation:** Incorporate genetic variation by applying mutation to a portion of the progeny's portfolio. Certain components (asset weights) of the offspring portfolios are randomly changed by mutation.
6. **Replacement:** Replace portfolios in the current population with the offspring generated from crossover and mutation. This step ensures that the population size remains constant.
7. **Termination:** For a predetermined number of generations or until the convergence conditions are satisfied, repeat steps 2 through 6.
8. **Final Solution:** Once the termination condition is met, select the portfolio with the highest fitness in the final population as the optimal solution:

$$p_{\text{optimal}} = \arg \max_{p_i \in P} f(p_i)$$

By iteratively applying selection, crossover, mutation, and replacement operations, the Genetic Algorithm searches for an optimal portfolio allocation that maximizes the fitness function, representing an optimized combination of asset weights that meets the specified investment objectives and constraints.

3.2.4 PROPOSED METHOD

Our proposed model aims to optimize portfolio allocation in commodity markets, focusing on five distinct commodity categories: Metals, Softs, Energy, Grains, and Meats, spanning from 2014 to 2023. The objective is to determine optimal weights for each commodity, maximizing return while minimizing volatility from January to February 2024.

Our proposed model integrates deep learning techniques with traditional portfolio optimization algorithms to enhance predictive accuracy and optimize portfolio performance. The initial step involves employing a CNN and LSTM layer. Commodity prices exhibit temporal dependencies, where past prices influence today's prices. LSTMs handle these dependencies effectively, allowing them to learn from historical data and make informed predictions. CNNs enhance this by extracting relevant features from the data, aiding the LSTM in capturing long-term patterns. Traditional recurrent neural networks (RNNs) suffer from the gradient vanishing problem, limiting their ability to learn long-term dependencies. LSTMs address this issue by using gates (input, forget, and output gates) to control information flow, preventing gradients from vanishing. CNNs can handle numerous input data channels. This combination enables the model to forecast average returns from January to February 2024 by efficiently capturing intricate temporal trends in the previous commodity price data. When optimistic return forecasts indicate potential profit opportunities, the model applies two traditional portfolio optimization algorithms: Hierarchical Risk Parity (HRP) and Genetic Algorithm (GA). HRP organizes the assets into a hierarchical structure based on their correlation, ensuring efficient risk allocation across different levels of the portfolio hierarchy. Meanwhile, GA iteratively evolves a population of candidate portfolios, selecting those that maximize return and minimize risk.

Following the application of HRP and GA, an optimization function is employed to refine the portfolio allocation further. This function aims to minimize the product of weights and volatility for each commodity, promoting diversification and effective risk management. Additionally, the optimization process incorporates constraints where the weights of each commodity are varied within a predefined domain, typically ± 0.03 , 0.04 , or 0.05 . Furthermore, the sum of all weights is constrained to equal 1 to maintain portfolio integrity. These constraints ensure the portfolio remains adaptable to market conditions while preventing excessive concentration in any single asset. Consequently, the model achieves enhanced stability and robustness by effectively balancing allocations across commodities within specified bounds.

Overall, this integrated approach combines the predictive power of deep learning with the robustness of traditional portfolio optimization techniques, resulting in a comprehensive framework for portfolio management in commodity markets.

In the context of our proposed model, a short-selling strategy is implemented when pessimistic return predictions are encountered for any commodity segment during the January to February 2024 period. Short selling is an investment strategy where an investor sells borrowed assets, with the expectation that their price will decline. The investor aims to buy back the assets at a lower price, return them to the lender, and pocket the difference as profit. Maintaining equal weightage across commodities ensures that each commodity has an equal impact on the portfolio's overall performance. This approach mitigates bias towards any specific commodity and promotes diversification, reducing the portfolio's overall risk exposure. The flexibility of weight adjustments within predefined ranges allows for dynamic adaptation to changing market conditions. By constraining weight adjustments to a specified neighborhood, such as ± 0.03 , 0.04 , or 0.05 , the model ensures that asset allocations remain moderate, preventing excessive exposure to any single commodity while allowing adjustments based on market dynamics. This adaptability enhances the model's ability to navigate volatile commodity markets and optimize portfolio performance in varying economic environments.

Comparison with established portfolio optimization methods such as Modern Port-

folio Theory, HRP, and HERC will validate the effectiveness and robustness of our proposed approach.

The following section offers an in-depth exposition and computational framework of the model.

CNN+LSTM MODEL

Historical data for the five commodity categories, namely METALS, SOFTS, ENERGY, GRAINS, and MEATS, spanning from 2014 to 2023, is collected. The data includes daily close, open, high, and low prices for each individual commodity within these categories. Financial elements like technical and fundamental indicators that examine the trends and prices of stocks can greatly improve the accuracy of forecasting algorithms [18]. Technical indicators are useful for characterising and depicting market circumstances, as previous research has shown. For example, Chen and Hao [13] suggest that one way to improve one’s comprehension of the stock market is to look at the correlations between technical indicators like the Momentum Index (MoM), Relative Strength Index (RSI), and Exponential Moving Average (EMA). Furthermore, True Range (TR), Average True Range (ATR), MoM, and RSI have been found by Wang et al. [19] to be relevant indicators for stock price forecasting, highlighting their usefulness in identifying market trends and behaviours.

1. Simple Moving Average (SMA):

$$\text{SMA}_t = \frac{1}{n} \sum_{i=0}^{n-1} \text{Close}_{t-i}$$

2. Exponential Moving Average (EMA):

$$\text{EMA}_t = \alpha \cdot \text{Close}_t + (1 - \alpha) \cdot \text{EMA}_{t-1}$$

where $\alpha = \frac{2}{N+1}$ and N is the number of periods.

3. Relative Strength Index (RSI):

$$RSI_t = 100 - \frac{100}{1 + RS_t}$$

where RS_t is the relative strength defined as the ratio of average gain over n periods to the average loss over the same n periods.

4. Rate of Change (ROC) or Momentum:

$$ROC_t = \frac{Close_t - Close_{t-n}}{Close_{t-n}} \times 100$$

5. True Range (TR):

$$TR_t = \max\{High_t - Low_t, |High_t - Close_{t-1}|, |Low_t - Close_{t-1}|\}$$

6. Average True Range (ATR):

$$ATR_t = \frac{1}{n} \sum_{i=1}^n TR_i$$

7. Momentum:

$$Momentum_t = Close_t - Close_{t-n}$$

Considering the time series nature of the dataset, understanding the evolving relationship between features and the outcome variable is essential. To identify relevant features, we employ Pearson's correlation coefficient, a widely used filtering technique in time series analysis. This method facilitates the identification of features with significant correlations with the target variable, aiding in the selection of pertinent predictors for analysis.

Features with a correlation coefficient greater than a predefined threshold ($r = 0.4$) with the close price are selected for further analysis.

Before feeding the data into the CNN+LSTM model, normalization is performed to scale the features. Normalization is a crucial preprocessing step before feeding data

into the CNN+LSTM model. Normalizing the data prevents features with larger magnitudes from dominating the model's training process. This is particularly important in cases where features have vastly different scales, as it helps prevent numerical instabilities and improves the convergence of the optimization algorithm during training.

Normalising the input data in the CNN+LSTM model for portfolio optimisation enables the algorithm to learn from past price movements and technical indications across many commodities, enabling more precise forecasts of future returns.

Now 20 percent of the data will be allocated for testing, leaving the remaining 80 percent for training.

The configuration of the model is described below.

Initially, the Conv1D layer applies filters with a kernel size to extract spatial patterns from the input sequences. This is followed by a MaxPooling1D layer, which reduces spatial dimensionality while retaining essential features through maximum pooling operations. To prevent overfitting, dropout regularization is applied after the convolutional and pooling layers. Subsequently, two stacked LSTM layers capture temporal dependencies within the sequences. These LSTM layers utilize the Rectified Linear Unit (ReLU) activation function to introduce non-linearity, enabling the model to learn complex temporal patterns. The output layer comprises a single neuron with linear activation, facilitating continuous value prediction. This hybrid architecture allows the model to effectively learn and predict from time series data.

The model predictive accuracy is assessed using six distinct performance metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), R-squared (R2), Maximum Error, and Median Absolute Error (MedAE).

1. **Mean Absolute Error (MAE):** MAE quantifies the average absolute difference between predicted and actual values, providing a straightforward measure of the model's accuracy. Lower MAE values indicate better performance.

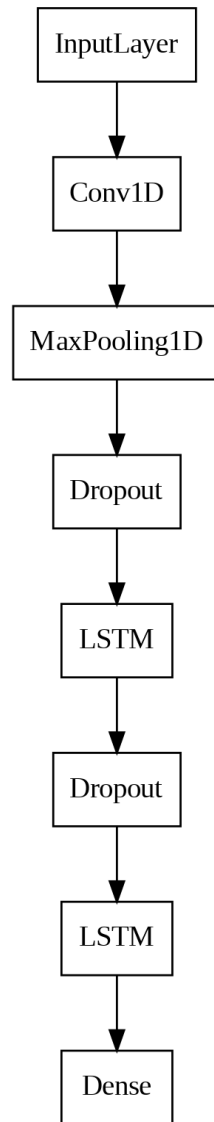


Figure 3.6: configuration

Model: "sequential"

Layer (type)	Output Shape	Param #
conv1d (Conv1D)	(None, 5, 64)	256
max_pooling1d (MaxPooling1D)	(None, 2, 64)	0
dropout (Dropout)	(None, 2, 64)	0
lstm (LSTM)	(None, 2, 50)	23000
dropout_1 (Dropout)	(None, 2, 50)	0
lstm_1 (LSTM)	(None, 50)	20200
dense (Dense)	(None, 1)	51
Total params: 43,507		
Trainable params: 43,507		
Non-trainable params: 0		

Figure 3.7: model summary

2. **Root Mean Squared Error (RMSE):** RMSE is akin to MAE but gives higher weightage to significant errors. It calculates the square root of the average squared differences between predicted and actual values, making it sensitive to outliers and helpful in assessing error dispersion.
3. **Mean Absolute Percentage Error (MAPE):** MAPE computes the average percentage difference between predicted and actual values, offering a relative measure of accuracy suitable for datasets with varying scales. Lower MAPE values signify better performance.
4. **R-squared (R2):** R2 elucidates the proportion of variance in the dependent variable (target) explained by the independent variables (predictions). Ranging from 0 to 1, higher R2 values denote a superior model fit to the data, with R2=1, indicating a perfect fit and R2=0 implying no explanatory power.
5. **Maximum Error:** Maximum Error identifies the most considerable absolute difference between predicted and actual values, providing insight into the model's worst-case performance scenario and aiding in outlier detection.
6. **Median Absolute Error (MedAE):** MedAE represents the median of absolute differences between predicted and actual values. It exhibits robustness against outliers, making it a reliable measure of central tendency in the error distribution.

These performance metrics collectively provide a comprehensive evaluation of the model's predictive capabilities, considering different aspects of accuracy, variability, and robustness.

The efficacy of the model is demonstrated for the Silver in Fig 3.8 and 3.9.

Loss function for train and validation data is shown in Fig 3.10,3.11,3.12

Now with this model average returns for the next two months i.e January and February 2024 is calculated for all commodities.

The table below shows the average predicted returns for all commodities in Fig 3.13,3.14,3.15,3.16,3.17.

Prediction from CNN+LSTM model

For Energy **Crude oil**, **Heating Oil**, and **Gasoline** are giving positive average returns for the next two months while **Natural Gas** is giving negative returns. So, we select Crude Oil, Heating Oil and Gasoline based on positive returns and then apply HRP and Genetic Algorithm individually. Similar is the case for Meats, where they all give positive average returns, so we selected all three. In the case of **Softs** except **Sugar** all of them are giving positive average returns over the next 2 months, so we select **Lumber**, **Orange**, **Cocoa**, **US Coffee C**, **US Cotton** and apply HRP and Genetic algorithm individually.

For Metals **Aluminium**, **Lead**, **Palladium**, **Tin**, **Zinc**, **Gold**, **Silver**, and **Nickel** all are giving negative average returns for the next two months, so we short sell this commodity. In case of **Grains**, except **Rough Rice** all are giving negative average returns for the next two months. So we select **Oats**, **US Corn**, **US Soybean Meal**, **US Soybean Oil**, **US Soybean**, and **US Wheat** and short sell this commodity.

In case of Metals we first assign equal weightage 0.125 to **Aluminium**, **Lead**, **Palladium**, **Tin**, **Zinc**, **Gold**, **Silver**, and **Nickel**. For Grains we allocate equal weightage 0.16 to **Oats**, **US Corn**, **US Soybean Meal**, **US Soybean Oil**, **US Soybean**, and **US Wheat**.

Volatility optimisation function

Volatility refers to the degree of variation or dispersion in the prices of a financial asset over time. It is a measure of the uncertainty or risk associated with the price movements of an asset. High volatility indicates that the asset's price can fluctuate dramatically in a short period, while low volatility suggests that the price changes are relatively stable and predictable. Here we have calculated the expected volatility of all the commodities for the next two months.

Objective function:

$$\text{Minimize: } \sum_{i=1}^n w_i \times \text{Volatility}_i$$

Constraints:

$$\begin{aligned} \text{subject to: } \quad & \sum_{i=1}^n w_i = 1 \\ & w_i - 0.03 \leq w_i \leq w_i + 0.03 \end{aligned}$$

Here, Volatility_i represents the volatility of commodity i over the upcoming two months, and w_i is the weight assigned to commodity i . The constraints ensure that the weights vary within the specified neighborhood of ± 0.03 while summing up to 1.

We similarly use this function for varying weights by 0.04 as well as 0.05 to find out the minimum volatility. After getting the weights which has minimum volatility, the portfolio overall variance is also reduced.

Training data metrics:
MAE: 0.20462853895585276
RMSE: 0.29668149933297966
MAPE: 0.010242632036641214
Max Error: 1.8987329101562516
MedAE: 0.14617530822754077
R2: 0.9937423590905303

Figure 3.8: Training

Testing data metrics:
MAE: 0.7153528988947635
RMSE: 0.7668126618930361
MAPE: 0.031121165703616883
Max Error: 1.488579177856444
MedAE: 0.6980229568481455
R2: 0.8631625663022607

Figure 3.9: Testing

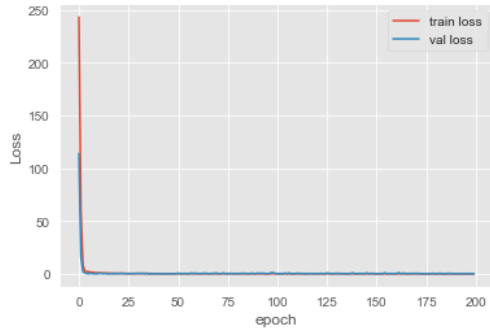


Figure 3.10: train loss and val loss

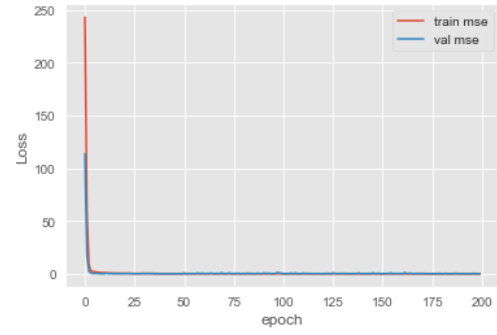


Figure 3.11: train mse and val mse

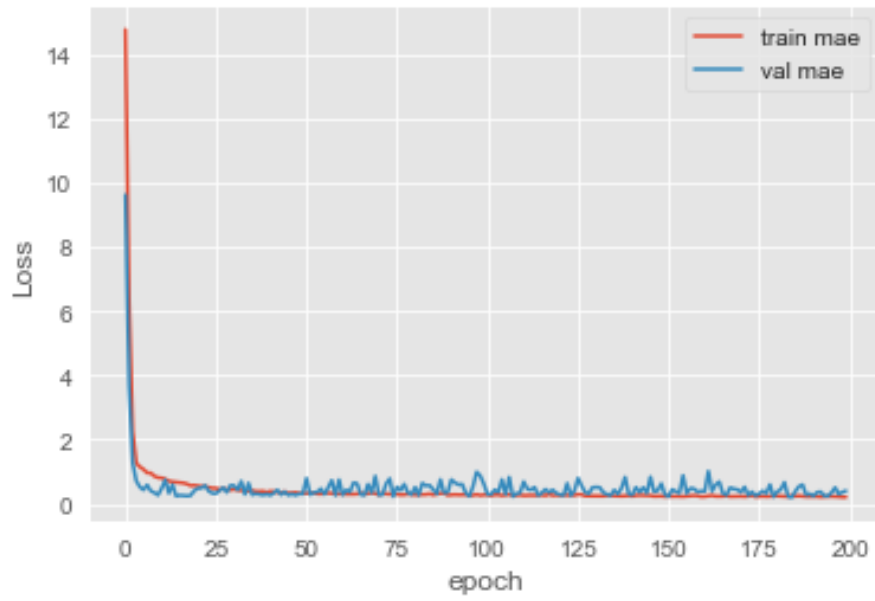


Figure 3.12: train mae and val mae

NAMES	R2 SCORE FOR TEST DATA	AVERAGE PREDICTED RETURNS
ALUIMINUM	0.85	-0.18%
LEAD	0.93	-0.04%
SILVER	0.83	-0.47%
PALLADIUM	0.96	-0.38%
GOLD	0.26	-0.20%
TIN	0.98	-0.73%
ZINC	0.9937	-0.17%
NICKEL	0.835	-0.40%

Figure 3.13: Average predicted returns for Metals over the next 2 months.

NAME	R2 SCORE FOR TEST DATA	AVERAGE PREDICTED RETURNS
CRUDE OIL	0.98	0.21%
HEATING OIL	0.69	0.83%
GASOLINE	0.64	0.19%
NATURAL GAS	0.82	-0.02%

Figure 3.14: Average predicted returns for Energy over the next 2 months.

NAME	R2 SCORE FOR TEST DATA	AVERAGE PREDICTED RETURNS
LUMBER	0.98	0.005%
ORANGE	0.97	0.14%
COCOA	0.97	0.32%
COTTON	0.90	0.09%
SUGAR	0.78	-0.84%
COFFEE	0.98	0.87%

Figure 3.15: Average predicted returns for Softs over the next 2 months.

NAME	R2 SCORE FOR TEST DATA	AVERAGE PREDICTED RETURNS
FEEDER CATTLE	0.97	0.60%
LEAN HOGS	0.98	0.58%
LIVE CATTLE	0.97	0.04%

Figure 3.16: Average predicted returns for Meats over the next 2 months.

NAMES	R2 SCORE FOR TEST DATA	AVERAGE PREDICTED RETURNS
OATS	0.94	-0.04%
ROUGH RICE	0.61	0.43%
CORN	0.97	-0.22%
SOYABEANS	0.62	-0.39%
SOYABEANS MEALS	0.65	-0.26%
SOYABEANS OIL	0.88	-0.53%
WHEAT	0.90	-0.60%

Figure 3.17: Average predicted returns for Grains over the next 2 months.

Chapter 4

Results and Discussion

4.1 Results

4.1.1 COMMODITIES GIVING POSITIVE AVERAGE RETURNS ENERGY

For Energy **Crude oil**, **Heating Oil**, and **Gasoline** are giving positive average returns for the next two months. We employ Genetic and HRP algorithm individually and then we got the separate weights from both. Then we apply our volatility optimisation function on the weights we got as described above. Final weights from Genetic were **Crude oil** (0.54), **Heating Oil** (0), and **Gasoline** (0.46). For HRP it was **Crude oil** (0.09), **Heating Oil** (0.87), and **Gasoline** (0.04).

The returns for two months from our method was 15.11% with annualised volatility of 19.78%. The maximum returns from MPT, HRP, HERC was 3.74%.

SOFTS

In the case of **Softs** except **Sugar** all of them are giving positive average returns over the next 2 months, so we select **Lumber**, **Orange**, **Cocoa**, **US Coffee C**, **US Cotton** and apply HRP and Genetic algorithm individually. After getting the weights we use our volatility optimisation function and final weights from

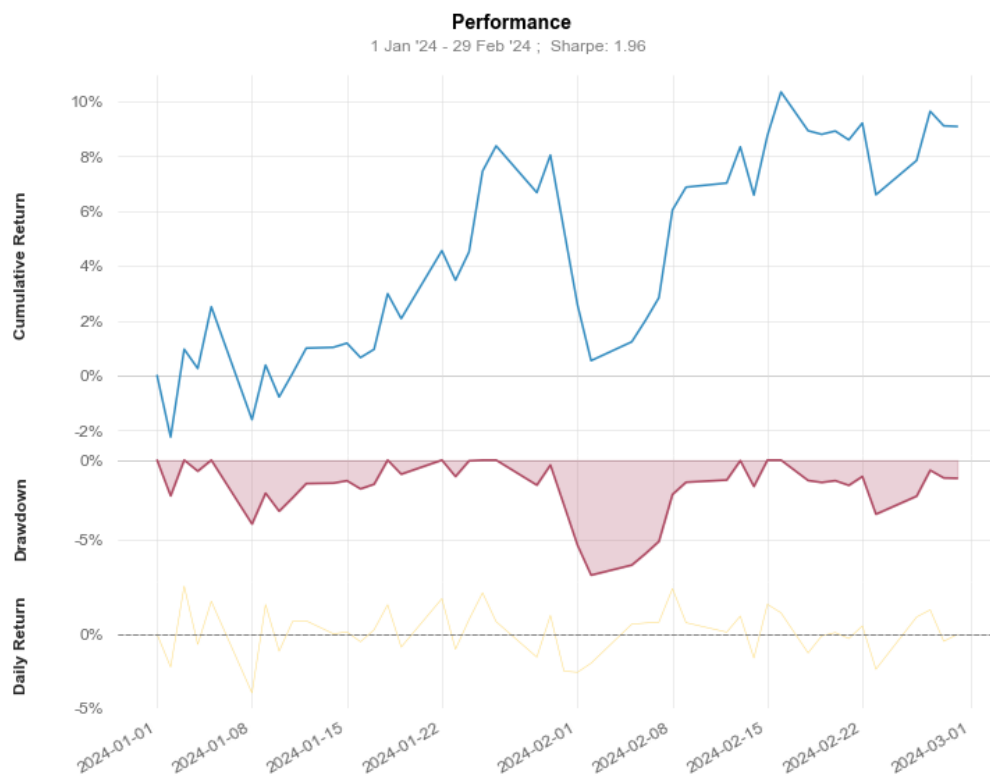


Figure 4.1: Cumulative Return ENERGY



Figure 4.2: Cumulative Return SOFTS

Genetic were **Lumber**(0.04), **Orange**(0.14), **Cocoa**(0.32), **US Coffee C**(0.18), **US Cotton**(0.32) .For HRP it was **Lumber**(0.02), **Orange**(0.12), **Cocoa**(0.34), **US Coffee C**(0.18), **US Cotton**(0.34).

The returns for two months from our method was 27.15% with annualised volatility of 39.42%.The maximum returns from MPT, HRP, HERC was 20.75

MEATS

Feeder Cattle, **Lean Hogs**, and **Live Cattle** all of them are giving positive average returns .we do the same process and the final weights for them from HRP were **Feeder Cattle**(0.46), **Lean Hogs**(0.09), and **Live Cattle**(0.45).From Genetic those were **Feeder Cattle**(0.53), **Lean Hogs**(0), and **Live Cattle**(0.47).

The returns for two months from our method was 11.58% with annualised volatility of 11.10%.The maximum returns from MPT, HRP, HERC was 10.5%.

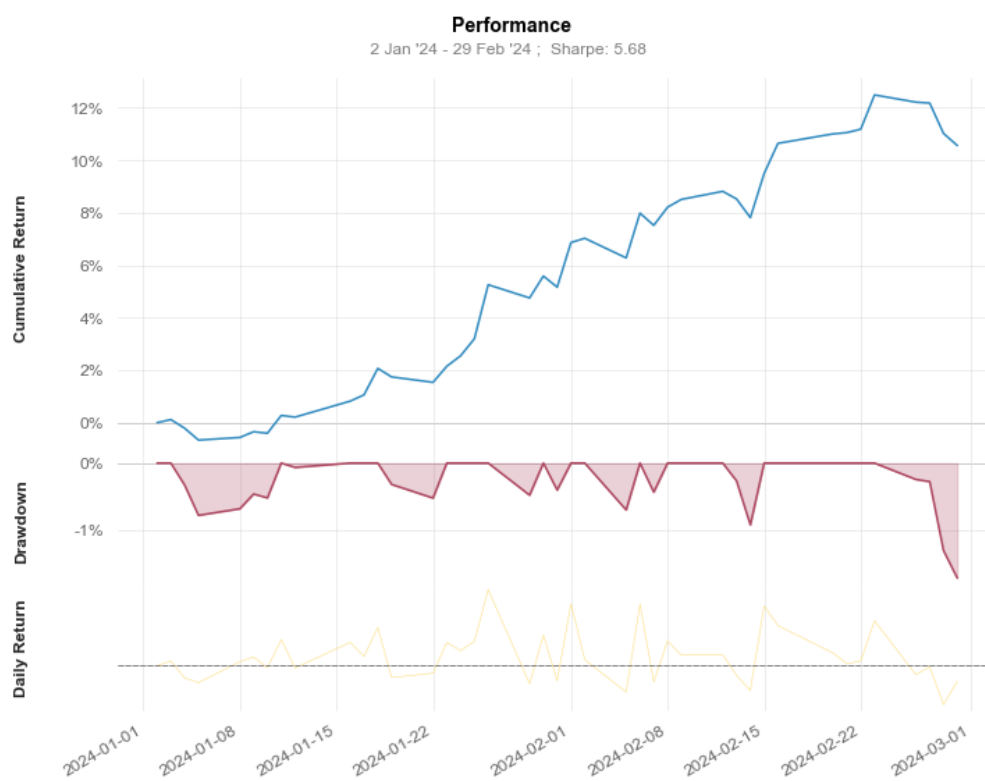


Figure 4.3: Cumulative Return MEATS

4.1.2 COMMODITIES GIVING NEGATIVE AVERAGE RETURNS

GRAINS

In case of **Grains**, except **Rough Rice** all are giving negative average returns for the next two months. So we select **Oats**, **US Corn**, **US Soybean Meal**, **US Soybean Oil**, **US Soybean**, and **US Wheat** and short sell this commodity. The final weights after applying the optimization function were **Oats** (0.11), **US Corn** (0.21), **US Soybean Meal** (0.15), **US Soybean Oil** (0.21), **US Soybean** (0.21), and **US Wheat** (0.11).

The returns for two months from our method was **8.37%** with annualised volatility of 12.34%. The maximum returns from MPT, HRP, HERC was **-5.63%**.

METALS

For Metals **Aluminium**, **Lead**, **Palladium**, **Tin**, **Zinc**, **Gold**, **Silver**, and **Nickel** all are giving negative average returns for the next two months, so we short sell this commodity. The final weights after the optimization function were **Aluminium** (0.15), **Lead** (0.15), **Palladium** (0.19), **Tin** (0.15), **Zinc** (0.13), **Gold** (0.15), **Silver** (0.09), and **Nickel** (0.09).

The returns for two months from our method was **2.38%** with annualised volatility of 13.87%. The maximum returns from MPT, HRP, HERC was **-1.37%**.

Chapter 5

Conclusion

In conclusion, this thesis has delved into portfolio optimization within commodity markets' dynamic and complex landscape. Through rigorous analysis and experimentation, we have devised a novel approach to strike a delicate balance between maximizing returns and minimizing volatility.

The efficacy of our approach was demonstrated through extensive comparisons with established portfolio optimization frameworks, including modern portfolio theory, HRP, and HERC. By surpassing these benchmarks in performance metrics and practical applicability, our method emerges as a promising contender in commodity market analysis and portfolio management.

Furthermore, incorporating adaptive weight adjustments, allowing for fine-tuning within specified bounds, enhances the robustness and flexibility of our model. This adaptive strategy, coupled with the prudent decision to short-sell underperforming segments, strengthens our commitment to optimizing risk-adjusted returns while mitigating downside exposure.

Chapter 6

Limitations and Future Work

In the pursuit of portfolio optimization within commodity markets, several limitations and areas for further research emerge. Firstly, while our proposed methodology shows promise in balancing returns and volatility, its effectiveness may be contingent upon the specific characteristics and behaviors of the commodities under consideration. As such, generalizing the approach across diverse commodity classes may necessitate further refinement and customization to account for varying market dynamics.

Additionally, the reliance on historical data for predictive modeling introduces inherent uncertainties and risks, particularly in volatile commodity markets.

Overall, while our research represents a significant step towards portfolio optimization in commodity markets, there remain ample opportunities for future exploration and innovation to address the inherent complexities and challenges of this dynamic and ever-evolving domain.

Chapter 7

Acknowledgment

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